

Real World AI Ethics for Data Scientists

Practical Case Studies

**Nachshon (Sean) Goltz
and Tracey Dowdeswell**

**In Collaboration with Peter Singer
and Yip Fai Tse**



CRC Press

Taylor & Francis Group

Boca Raton London New York

CRC Press is an imprint of the
Taylor & Francis Group, an **informa** business
A CHAPMAN & HALL BOOK

Designed cover image: © Getty Images

First edition published 2023

by CRC Press

6000 Broken Sound Parkway NW, Suite 300, Boca Raton, FL 33487-2742

and by CRC Press

4 Park Square, Milton Park, Abingdon, Oxon, OX14 4RN

CRC Press is an imprint of Taylor & Francis Group, LLC

© 2023 selection and editorial matter, Nachshon (Sean) Goltz and Tracey Dowdeswell; individual chapters, the contributors

Reasonable efforts have been made to publish reliable data and information, but the author and publisher cannot assume responsibility for the validity of all materials or the consequences of their use. The authors and publishers have attempted to trace the copyright holders of all material reproduced in this publication and apologize to copyright holders if permission to publish in this form has not been obtained. If any copyright material has not been acknowledged please write and let us know so we may rectify in any future reprint.

Except as permitted under U.S. Copyright Law, no part of this book may be reprinted, reproduced, transmitted, or utilized in any form by any electronic, mechanical, or other means, now known or hereafter invented, including photocopying, microfilming, and recording, or in any information storage or retrieval system, without written permission from the publishers.

For permission to photocopy or use material electronically from this work, access www.copyright.com or contact the Copyright Clearance Center, Inc. (CCC), 222 Rosewood Drive, Danvers, MA 01923, 978-750-8400. For works that are not available on CCC please contact mpkbookspermissions@tandf.co.uk

Trademark notice: Product or corporate names may be trademarks or registered trademarks and are used only for identification and explanation without intent to infringe.

ISBN: 978-1-032-27506-2 (hbk)

ISBN: 978-1-032-27505-5 (pbk)

ISBN: 978-1-003-29312-5 (ebk)

DOI: 10.1201/9781003293125

Typeset in Bembo

by KnowledgeWorks Global Ltd.

2 Introduction to Ethical Approaches in Data Science

Knowledge of physical science will not console me for ignorance of morality in time of affliction, but knowledge of morality will always console me for ignorance of physical science.

~ (Blaise Pascal 1624–1663)

Introduction

Machine learning technologies are encroaching into the lives of ordinary people around the world. Users of these technologies – some willingly, some unwillingly – have ethical lives that are informed by approaches not represented among Western philosophies. This is the lens through which these users of technologies come to understand them.¹ We have therefore included several non-Western approaches to ethics. These are worth knowing for designers, both so that they can deepen their own exploration of ethics, and so that they can better understand how their technologies are going to be interpreted, adopted, used, and (inevitably) regulated.

We have been fortunate to receive commentaries from leading scholars in the fields of deontological ethics, consequentialist ethics, and virtue ethics, as well as from Ubuntu ethics, Buddhist ethics, Jewish ethics, and Indigenous ethics. We hope that this will give the reader a broader base to begin to think about machine learning technologies from different perspectives and to understand how they will be adopted by and function in societies around the world. Each of these ethical approaches is briefly summarized below.

Consequentialism and Utilitarianism

By Peter Singer and Yip Fai Tse

Consequentialism is a family of theories that hold the view that whether an action is right or wrong depends on its consequences, or in other words, the states of affairs the actions cause. Utilitarianism, in its classical form, is the consequentialist theory that focuses exclusively on pain and pleasure, or happiness and misery, as the only morally relevant consequences for determining how the consequences of actions are to be evaluated.

Here, it is important to emphasize that utilitarianism is not only about assessing the rightness or wrongness of actions, but also about assessing the goodness and badness of states of affairs, considered impartially. Specifically, Utilitarians hold that all sentient beings – those that can experience pain and pleasure – must be considered, and their similar interests² should be given similar weight. Put together, Utilitarianism is therefore the view that an action is not only right but also morally required when it brings about the greatest possible net surplus of happiness over misery (or of pleasure over pain); and also that an action is wrong and impermissible when it does otherwise.

Common Objections to Utilitarianism

A common objection to Utilitarianism is that it directs us to perform clearly immoral acts. In *The Brothers Karamazov*,³ Dostoevsky has Ivan challenge his brother Alyosha to imagine whether he would accept torturing a baby to death in order to bring happiness to all of humanity. Ivan's challenge has become a famous objection to Utilitarianism. Setting out the structure of Dostoevsky's objection formally better illustrates the issue:

- Premise 1. If Utilitarianism were true, it would tell us, correctly, which acts are right and which are wrong.
- Premise 2. Utilitarianism tells us that if torturing an innocent child to death would bring about better consequences than any other action, then torturing an innocent child to death would be the right thing to do.
- Premise 3. Torturing an innocent child to death is always wrong.
- Conclusion: Utilitarianism is false.

Many objections to Utilitarianism are similarly constructed: a surgeon contemplates whether to secretly ensure that an operation fails so that the patient dies and her organs can be used to save the lives of four patients waiting for donations of essential organs. Such examples do not reflect our knowledge of how the world works. Ivan did not explain how torturing the child would bring lasting happiness to everyone else. The organ transplant example fails to consider that if what the surgeon has done becomes known, this might lead to consequences that far outweigh the intended benefits – people might become distrustful of doctors and reluctant to seek medical help. How can the surgeon be entirely sure that she would not be caught? The assumption that torturing an innocent child is always wrong relies on an intuition that evolved to respond to circumstances we are likely to encounter. When we are dealing with fantastic examples, Premise 3 is therefore doubtful and cannot be relied upon as a basis for refuting utilitarianism.

Another major objection is that pain and pleasure, or welfare states, are hard to measure. Utilitarians have three main responses to this objection. Firstly, this is not a problem limited to Utilitarianism. Any moral theory that gives some weight to welfare will also suffer from the difficulty of measuring

the welfare of those affected by our actions – and a moral theory that ignored all such welfare considerations would be extremely implausible.

Secondly, even though it is difficult to measure pain and pleasure precisely, the preferences of people, and to some extent animals, can be observed, tested, and ranked to reveal their priorities. In some studies, psychologists test people's willingness to pay to avoid⁴ or endure⁵ certain levels of pain. These revealed preferences, although not what classical Utilitarians regard as the ultimate good, are nevertheless useful proxies that give us some idea about the underlying pain and pleasure.⁶ Another model using revealed preferences, the quality-adjusted life-year, or QALY, is framed around the idea that a year of life with impaired functionality or health is not as good as a year in normal health. To find out how much less good it is, researchers ask people to imagine themselves with various impairments to their health (sometimes pain itself), and then ask them how many years of life they would give up in order to have that impairment cured. This method is now used globally by health economists, medical researchers, and policy-makers.⁷

Finally, in many cases, the right action is clear even without measurement. For example, a doctor prioritizing the order of patients can clearly see that a heavily burned patient is in much more pain and at much higher risk of dying than someone suffering from a cold, and should therefore prioritize the burn patient. Or, if someone asks you how to get to the nearest vegan restaurant and you know the answer, you will very likely maximize utility by providing her with the information she is seeking, rather than not responding at all, or lying to her. In both cases no “measurement” is needed.

While it is true that there are cases where things are not nearly as clear, after carefully analyzing evidence, reasonable choices can nevertheless be made. An important point to consider here is that not only can a significant portion of *decisions* under Utilitarianism be made without measuring pleasure and pain, but what is *at stake* in this world can also usually be determined without direct measurements of pain and pleasure. Global poverty (which causes hunger, thirst, diseases, and “poverty traps”), factory farming, and pandemics are pertinent examples of issues that undoubtedly cause enormous amounts of suffering to an enormous number of individuals.

Advice for Properly Applying Utilitarian Principles

Think Broader and Longer

We agree with John Stuart Mill, an early utilitarian, that we should “regard utility as the ultimate appeal on all ethical questions; but it must be utility in the largest sense”. By “largest”, we mean that all relevant consequences should be considered, regardless of their timing, physical distance, kinship, and other morally irrelevant characteristics such as gender, race, and species membership.

Arguably, time is one of, if not the, most controversial of these to be declared morally irrelevant. Time discounting is often taught and applied in the fields of economics and machine learning, and their notions about time preferences are different from those of Utilitarianism. In economics, time discounting is used to capture people's general preferences for a nearer enjoyment than a farther one. In machine learning, particularly in reinforcement learning, the "discount factor" (or γ) is the variable that determines how much a reinforcement learning agent discounts rewards depending on how far in the future they are. It is used for various reasons: it prevents the reward from becoming infinite, it captures the intuition that there is generally more uncertainty in the future, and it sometimes improves the stability of the agent.⁸ These practices of using discounts might give one the impression that time preference is justified. But we need to distinguish between pure time preference due solely to time itself and discounting time for other reasons. Utilitarians, because of their impartiality, do not hold pure time preferences but accept that other practical reasons, such as uncertainty, are justified reasons for discounting. For example, torture in 100 years is equally bad as torture that brings the same amount of pain now, but if it is less certain, we may discount it for that reason.

Let us try to apply these principles to artificial intelligence (AI) and data science. In deciding to launch a product, for example, one should not only consider the impact it might have on its users, but also how the broader society of individuals (in some cases, even nonhuman animals) could be affected, both in the short-term (such as within the product's lifecycle) and in the long-term. Questions such as these should be asked: Will this product capture, and thereby reinforce biases, culture, ideologies, virtues, or other values in society? Will this product eliminate a hugely valuable industry, or will it delay or prevent the elimination of a morally objectionable industry?

Use Expected Values to Make Decisions

Using expected value theory in decision-making is foundational in decision theory, economics, and data science. But it should also be applied to ethical theories, especially maximizing ethical theories such as Utilitarianism. The surgeon example in the previous section shows why low-probability-high-stakes scenarios are important. No matter how carefully the surgeon tried to keep her act secret, she could not reasonably conclude that the chance of the secret being exposed is zero. Considering the potentially catastrophic effects of being discovered, the surgeon ought to conclude that it would be wrong to perform such an act.

While calculating expected value is often straightforward, it is often improperly applied or even not applied at all, due to human cognitive biases. Probability neglect is a cognitive bias people show toward uncertainties, especially that of small probabilities, which they have the tendency to either neglect entirely or exaggerate greatly.⁹ A study confirmed this by finding that

people placed either a high or a very low value on mitigating risks of “rare, high-impact events”.¹⁰ We need to look no further for evidence of a collective neglect of “rare, high-impact events” than the fact that without legal compulsion, many drivers and passengers in cars would not do up their seat belts. This pitfall in human thinking is worrying, as events with low probability and the high impact often have large expected values, either negative or positive.

Another bias that might further affect people’s ability to estimate the expected values is scope neglect. Studies found that people do not adjust their valuation of an issue in proportion to its scale.¹¹ For example, one study asked three groups of subjects about their willingness to pay to save 2,000, or 20,000, or 200,000 birds from drowning in uncovered oil ponds. The respective means were \$80, \$78, and \$88, and the median answers were all \$25.¹² If people’s valuations of certain outcomes do not scale properly, the expected values won’t either.

In Choosing Charitable Projects, Choose Effective Ones

As people commonly give to charity on the basis of impulse and emotion, rather than research into the effectiveness of the charity, they often support ineffective charities and causes.¹³ In response, Effective Altruism, a recent global movement, emphasizes the importance of effective altruistic behavior, either in the form of monetary donations or time. One of the movement’s most important contributions has been to assess the available evidence in order to discover and publicize the causes and organizations that are most highly effective in doing good.

Applying the same principle to the fields of AI and data science, companies and scientists searching for ways to do good should prioritize the approaches that show evidence of being highly effective or having high expected value. Examples of this research, and sample analyses on the effectiveness of various causes, careers, and charitable programs can be found on the websites of such organizations as Giving What We Can, 80,000 Hours, The Life You Can Save, and Animal Charity Evaluators.¹⁴

Deontological Ethics

By Colin Marshall

Deontological approaches to ethics focus on a cluster of related ideas: respect, autonomy, rights, and refusing to treat humans (and perhaps other beings) as though they were mere things or means to other ends.

A classic illustration of the deontological approach involves the following scenario: a doctor has five patients each in urgent need of organ transplants. A trusting and healthy person walks into the doctor’s office, and the doctor could kill the healthy person and harvest their organs in order to save the five

patients. Even if the doctor could do this without being detected, many people judge they should not. This judgment is easily expressed in deontological terms: as a failure of respect on the doctor's part, as a violation of the healthy person's rights, or as the doctor using the healthy person as though they were a mere a thing (an organ container).

The deontological approach is often contrasted with consequentialist approaches, which recommend whichever action yields the best outcome. And, if the details were filled out the right way, a consequentialist approach to the doctor case could recommend that the doctor sacrifice the one healthy person to save the five patients. In practice, however, the verdicts of deontological and consequentialist ethical approaches frequently coincide. After all, in any *realistic* version of the doctor case, there could be no guarantee that the murder would remain secret. Given that, consequentialism would recommend factoring in the huge risks involved, such as decreased trust in medical professionals (resulting in sick people failing to seek necessary help) and the likely devastating psychological impact on the doctor (whose guilt and trauma might detract from their future work). As a result of such considerations, many consequentialists have held that the best consequences would be secured if people generally approached decision-making from a deontological perspective. For that reason, we can expect many deontological evaluations of ethical cases to agree with consequentialist (and other) evaluations, even if the different approaches emphasize different factors.

In the commentary below, the central deontological notion will be that of *respect*, together with two notions that emerge from respect: *impartiality* and the *refusal to others as mere means*. We can briefly consider each of these here.

There are many varieties of respect, but the relevant form of moral respect is giving serious consideration to others' needs and projects. Such moral respect can and should often impact action: if we seriously consider someone's needs, we would typically not act so as to undermine those needs. Yet there can be failures of respect even when no action results, as in disrespectfully laughing at others' failures.

Respect is hardly our default attitude toward others. Instead, we are frequently drawn toward disrespectful partiality, focusing on our own needs and projects (or those of some group to which we belong) at the expense of others' needs and projects. For that reason, respect calls for impartiality in our dealings with others. It is difficult to tell when we are being appropriately impartial since we often construct impartial-sounding rationalizations for selfish behavior (consider the dangerously flexible cliché, "our product will make the world a better place"). For that reason, deontology often recommends certain forms of reflection to help detect partiality – see the commentary on Case 2 – "Machine Models in Court" for one example.

While impartial respect might be an absolute standard, partiality comes in degrees. Someone can be partial toward their social group, for example, while still giving *some* consideration toward other social groups. At an extreme, however, partiality involves giving *no* serious consideration toward

certain other beings' needs and projects, and so treating them as nothing more than tools for achieving certain goals. If a social media platform, with the sole aim of maximizing profit, entices its users into forms of harmful engagement, they are treating their users as mere means for achieving profit (see the commentary on Case 6 – "Mental Malware" for more). Similarly, if a factory farm treats animals as mere sources of meat, then it is treating them as mere means (see the commentary on Case 7 – "Animals and AI"). Such an attitude constitutes a complete failure of respect.

Whether or not deontology captures the ultimate ethical truth in some sense, many people care about respect, rights, and impartiality. There are therefore both economic as well as moral reasons for taking the deontological approach seriously.

Virtue Ethics

By John Hacker-Wright

Virtue ethics is an approach to ethics or, more accurately, a family of approaches, that grounds its guidance of action on the states of character that a human being needs to live well. It tells us to develop and exhibit good states of the character called virtues, and to avoid developing and exhibiting bad states of the character called vices. The most prominent strand of virtue ethics in the Western academy today is informed by the ancient Greek philosopher Aristotle (384–322 BCE), but there were and continue to be many versions of virtue ethics. Hence, one can find, for example, Confucian and Buddhist versions of virtue ethics. The view I will expound on what follows and that informs my approach to the cases is Aristotelian virtue ethics.

When thinking of a good person, you might think of someone with qualities such as courage, compassion, honesty, and the like. These are putative virtues. Whatever qualities we think someone needs to live well in a certain domain of human life comprise our conception of the virtues. While there is no definitive list of virtues, there is significant convergence over qualities such as courage, honesty, justice, and wisdom. Virtue ethicists try to derive a criterion of right and wrong in action from the virtues or the virtuous person. One prominent formulation goes: an action is right if and only if it is what a virtuous person, acting in character, would do in that situation.¹⁵ Note that we can follow this even if we are not, ourselves, virtuous, provided we have some level of insight into what the virtuous agent would do and sufficient self-control to carry through with acting as the virtuous person would act. If our desires are too disordered, we may not be able to carry through with virtuous intentions and may even act worse as a result of trying to act as a virtuous agent would act!¹⁶ In that case, our moral agency is compromised by the weakness of will, and a virtuous person would advise a remedial course of action. Our goal remains to become able to act as the virtuous agent would act.

Perhaps the most significant difference between virtue ethics and other approaches to ethics is that virtue ethics embraces non-codifiability: there are no exceptionless rules that could capture what it is to act well across the variety of situations we face in human life. There may be rules that capture loose generalities, patterns of action, and reasoning characteristics of virtuous people, but these cannot be applied unthinkingly. In other words, some level of moral understanding is necessary to apply them. This might seem like a weakness of the theory, but on the other hand, competing theories commit themselves to deeply counterintuitive and sometimes morally disturbing views of right action based on exceptionless rules: for example, Immanuel Kant's deontological view is infamously committed to an exceptionless position on never telling a lie, even if it would save lives to do so. Virtue ethicists, by contrast, may hold that the human need for trusting relationships makes honesty a virtue and yet contend that we can uphold our commitment to honesty while allowing circumstances that are calling for deception. For example, it may be appropriate to lie when we are pressed for information about someone's whereabouts by someone with obvious murderous intentions. The absence of exceptionless rules may also be an asset for virtue ethics when it comes to dealing with emergent technologies, which change the nature of our practical lives.¹⁷

Since the virtues are at the center of virtue ethics, it is crucial to understand what they are. Some virtues are excellences of our desires and feelings, whereas others, such as practical wisdom, are primarily intellectual excellences. Courage, for instance, concerns our desire for safety and it is exhibited when our feelings of fear and confidence are such that we feel fear only in the face of what is genuinely dangerous and to the extent that it is dangerous. Aristotle explained the idea of virtue by appealing to his famous "doctrine of the mean."¹⁸ In a courageous human being, the feelings of fear and confidence are in a medial state between an excess and a deficiency. Someone who feels excessive fear will flee from danger and fail to achieve something worthwhile. We label such people cowards as they exhibit the vice of cowardice.

Someone who feels too little fear may act recklessly and face injury or death in vain efforts that should have been avoided. It is characteristic of the Aristotelian approach that fear, among other feelings, is something that is necessary to live well; after all, when I feel fear, I register the value of my own life and physical integrity in a way that motivates action. Still, I may overvalue my life and physical integrity.

In Aristotle's view, there are more important things than my own life and physical integrity, such as the liberty of my city-state and the safety of my friends and family. Hence, on his view, it is a good thing to risk death if there is a non-trivial chance of achieving some such end. Another aspect of Aristotle's view is that one cannot exhibit courage unless one is facing fear for the sake of a worthwhile end. A thief who faces danger for the sake of stealing is not courageous. Although their character is such that they are not prone to feeling excessive fear, still, this state of character is not an excellence in them.

Their viciousness in another domain – say, acquisitiveness – undermines their ability to act courageously. This is another aspect of Aristotle’s doctrine: he advocates an idea called “the unity of the virtues” which states in its strongest form that to have one virtue we must have them all. In a somewhat weaker and more plausible form, the idea is that any vice undermines the ability to exhibit any virtue. Assuming that there are states intermediary between virtue and vice, this leaves room for being less than fully virtuous in some areas without necessarily undermining our virtue in other areas.

Staying with the virtue of courage as an example, we can wonder whether it is good to be courageous. After all, if it requires that I risk my life for the sake of my city-state, perhaps it is better for me to be cowardly. But note that this accepts the cowardly perspective on the world: that it is better at any cost to survive. The courageous person sees the world differently: survival is not good when it comes at the cost of the freedom of one’s city-state or the death or enslavement of one’s friends and family. Are we left, then, with the juxtaposition of these two different perspectives, or is there some objectivity to the courageous person’s perspective? I believe that the courageous person’s perspective is superior because courage is a trait that human beings need to live well in a world of danger. We human beings need to be able to carry through with goals even when faced with dangers. This view is a version of ethical naturalism that many Aristotelians embrace: that goodness in humans is a function of the sort of animal that they are.¹⁹ The virtues perfect human rational and appetitive powers, and it is an objective matter, of which traits do that.

Aristotle wrote in an age with a very different social structure as well as with different technologies. One might wonder whether his views are relevant to our times. Certainly, no Aristotelian virtue ethicist writing today accepts his views without modification. There is an over-emphasis on martial virtue of courage and a heavy aristocratic bias to his conception of the virtues; his political views are tainted with the misogyny and racism of his time. But his philosophical framework nevertheless still yields insight. In addressing questions of technology and data science, Aristotelian virtue ethics emphasizes an examination of the impact of the virtue on our character: how does the use of a new technology impact our desires and thinking? If a technology is making us think or feel something as characteristic of a vicious agent, then this is grounds for ethical criticism of the technology. Hence, the focus is on how well we are living with the technology. We do not have to find a dramatic impact on society or the violation of duties to raise moral qualms about a technology; we can approach ethical criticism of technology by taking stock of distortions and its effects on our thoughts and feelings. New technologies may make new ethical demands on us. In such cases, it is a question whether a new virtue is required, or whether it is simply a matter of thinking about a traditional virtue within a new context. My view is that new labels tend to repackage aspects of traditional virtues, and that there is no harm and may be some benefit in doing that, since it may help us to think

more carefully about the situations we face. In short, Aristotelian virtue ethics provides a flexible framework for thinking about how well we are living with new technologies, and it need not be tightly tied to Aristotle's ancient views about the virtues.

One final aspect of Aristotelian virtue ethics that is important to bear in mind is the idea, found in Aristotle, that knowledge of virtue and the conditions of its development to be important to politics. Virtue ethics is misunderstood if it is assumed that we as individuals are on our own to develop and exhibit traits that we need to live well under whatever circumstances. Instead, virtue ethics is or ought to be concerned with weighing in on the social conditions that are necessary for human beings to live well. This is especially important in considering the impact of new technologies. They may undermine (or, to be more optimistic, enhance) our ability to bring our desires into line with our conscious goals, and thereby undermine our efforts to develop the virtues. On the Aristotelian view, the development of virtues requires a process of habituation, that is, a process of acting as the virtuous agent would act, perhaps against our inclinations, until we come to enjoy acting in that way and can do it reliably on our own. This process happens in a community and begins before we are capable of conscious, intentional action. Hence, it matters very much what sort of community we start from, and much of what virtue ethics will have to say about technologies will come from examining the impact of technology on communities from a standpoint of whether it enables us to develop self-control and excellences of character.

A Provisional Perspective on African Ethics

By John Murungi

In what follows, one should expect a provisional perspective on African ethics. Provisionality matters as it leaves room for other perspectives. Moreover, it alerts the reader to the fact that what is said about African ethics does not exhaust what can be said about it; there is more to be said. African ethics lacks the solidity of a granite rock. Even such a rock, regardless of how hard it is to penetrate, remains penetrable, since nothing in nature is essentially impenetrable. What is true of African ethics is equally true of any other branch of ethics.

African ethics is not a datum placed on a calabash for warehousing or for transmission from one person to the next. It is not placed there for contemplation or for reflection by any one person. It is in the process of being what it is. One should not exclusively rely on an empirical perspective unless such a perspective is understood as provisional. The prejudice of empiricism solidifies data so that it can be subject to an empirical method of verification. This prejudice robs data of its essential processional and provisional aspect.

Aimé Césaire, a Caribbean of African descent from Martinique, was a member of the French Communist Party. In protest against excluding racism

as a fundamental category in the struggle against exploitation and oppression, he wrote a resignation letter in which he said, "... there are two ways to lose oneself: walled segregation in the particular or dilution in the 'universal'."²⁰ This is an important insight if one is to do justice to the understanding of African ethics. African ethics is a singular ethics as is the case with any other branch of ethics. It should not be mistaken for or confused with any other branch of ethics. It has indestructible autonomy. It must be understood in its own terms. It ought not to be diluted in the "universal." Inescapably and at the same time, it is not walled in segregation in the particular. It is universal, not in a false sense of universality but in true universality. True universality does not dispense with true particularity. It dispenses with false particularity. It is instantiated in the true particularity. True particularity is itself instantiated in true universality. The distinction between true universality and true particularity is an ongoing fundamental problem. The problem is intrinsic in what is honestly said about African ethics.

In speaking about African ethics, one must address what is African about ethics. What is African about African ethics is not exclusively a matter of physical geography or a matter of biology. If this was not the case, African ethics would be a matter of physical geographers or a matter of biologists. When ethics is at stake, one must pay attention to geo-ethicists or to bioethicists bearing in mind that ethics is not a physical science or a bio-physical science. Ethics is elemental and does not depend on what is extra-ethics for its being. What is African about African ethics has an irreducible aspect of true particularity and, at the same time, it has an inescapable aspect of true universality.

I propose that the particular-universal aspect of African ethics lies in the African concept of Ubuntu. This is claimed not at the expense of other concepts of African ethics – both existing and the ones to come. Ethics, whether African or non-African, whether particular or universal, is about well-being. In Indigenous African societies, societal well-being is communal well-being. It is well-being that has a place for individual well-being as well as group well-being. In either case, such well-being is not at the expense of society's well-being. This broader sense of societal well-being rests on the African concept of Ubuntu or on other similar concepts such as Ujamaa, or on what used to be referred to as African Socialism. In the Ubuntu mode of human being, it is claimed that "We are, therefore I am." In Ubuntu, it is recognized that, in the pursuit of well-being, there is a place for an individual or a group to pursue well-being but not at the expense of the pursuit of the well-being of society. It is also recognized that, in pursuit of well-being, an individual or a group could act in a manner that is contrary to societal well-being. In such cases, there are societal controls to minimize the threat to society's well-being. The legitimate pursuit of society's well-being, however, cannot be undertaken at the expense of legitimate individual or legitimate group pursuits. Legitimacy in either pursuit is a matter of ongoing deliberation. Moreover, borders of the individual, group,

or society are permeable. They are subject to change, and they too call for ongoing deliberation.

It is important to remind ourselves that African ethics is not subject to ethnography, or ethnology. It is not ethnic or tribal ethics. Today, in the light of colonization and racism, it is of utmost importance deracialize and to decolonize the concept of African ethics. If there is going to be an honest and a true version of African ethics, decolonization of the language of this ethics, both within and outside Africa, is necessary. It must not be forgotten that African ethics is what it is: African ethics. African ethics ought to be defined ethically. It is a human ethics – a human ethics defined ethically. It must be borne in mind that the process of definition is essentially incomplete, and, in the foreseeable future, it remains as such. Moreover, it should not be restricted to the domain of the human. The domain of the human extends to the domain of the non-human. It is what it is by being encompassed by the domain of all that is. Its frontiers are ever expanding. It is the acceptance of membership of Africans in this broader domain that lead European explorers in modern history to the radically erroneous belief that Africans were savages or uncivilized. Correcting this error is an essential aspect of African ethics.

Because ethics is implicated in the well-being of society, it seems proper that sociology is indispensable in the study of ethics. As sociology is the study of society, the study of ethics is embedded in sociology. Moreover, since society is politically secured and its interests are promulgated and pursued by the state (the political), ethics is fundamentally political. Differently stated, ethics is subject to political sociology. In conventional Euro-Western ethics, the multilayered architecture of ethics is rarely recognized. In the Indigenous African context, this architecture is recognized. The sense of the political in Africa is radically different from the modern European concept of the political. In the African context, the politico-ethical is akin to the Ancient Greek political sociology. In his book on politics,²¹ Aristotle tells us that by nature man is a political animal. Modern Euro-Western sense of the political deviates from this conception of the political. It seems that Indigenous Africans share the Greek sense of the political and, indeed, there are other societies that share it. Euro-Western modernity appears to be in flight from this political ethical sense. It is subversive of the ethical of the Indigenous Africa. To this extent, the Euro-Western sense of the ethical is in conflict with Indigenous African sense of the ethical.

Buddhist Ethics

By Peter Herschok

Ethics covers a considerable amount of conceptual ground. It has been equated with moral reflection on how 'we' do or do not do things from within a historically, culturally, and geographically located community. But it has also

been equated with rationally justified derivations of principles of conduct deemed applicable across and among communities, not just within them. Thus, ethics can involve everything from explications of what is involved ideally in being a “good” person, to that of the practical meaning of “acceptable” agency within a profession or as citizens of a nation or the world.

I approach ethics operationally, defining it minimally as the *evaluative art of human course correction*: the art of intelligently enacting the results of discriminating collaboratively and qualitatively among our values, aims, and interests, and our means for realizing them. For me, it is an art crucially informed by contemporary elaborations of Buddhist concepts and practices.

Buddhism emerged in the Himalayan foothills of South Asia some 2,600 years ago, at roughly the same time as the philosophical and political traditions of the Mediterranean and Sinitic worlds. Those traditions grappled with such fundamental questions as: what is real? what is good? what is the place of humanity in the cosmos? and, how should society be governed? Buddhism arose in therapeutic – rather than theoretic – response to two different, but equally fundamental questions. What are the causes and conditions for experiencing *duḥkha* or suffering, conflict, and trouble? And, by what means can we dissolve these causes and conditions?

Buddhist responses to these questions rest on two key insights. First, all things arise and persist interdependently. Strongly stated, relationality is more basic than things related. All things are a function of relational differentiation, and each thing ultimately *is* what it *means* to and for all others. Second, our cosmos is self-organizing and karmically-structured. It is a cosmos in which consistent patterns of values, intentions, and actions result in consonant patterns of experiential outcomes and opportunities.

The Buddhist art of human course correction has the aim of realizing freedom from *duḥkha*-engendering relational entanglements, most proximally through resolving conflicts among our values, intentions, and actions. This depends on critically assessing habits of thought, speech, and conduct, and on realizing the freedom of attention and freedom of intention required to revise, resist, or dissolve those habits as needed to be no longer constrained by karmic entanglements and compulsive presence. Significantly, the aim of Buddhist practice – realizing nirvana – is not prescribed or positively defined. Instead, it has traditionally been glossed metaphorically as cooling down or extinguishing the fires of craving, aversion, and ignorance. This has important ramifications for Buddhist ethics.

Among the simplest formulations of Buddhist practice is that, for the purpose of ending conflict, trouble, and suffering, one should see all things as impermanent and without fixed essence or identity, and as liable to triggering relational blockages and entanglements if seen otherwise. This practice opens possibilities for being present and acting in ways that are *kuśala* and eschewing those that are *akuśala*. Although it is often translated as “wholesome” or “skillful,” *kuśala* carries the normative force of a superlative. It directs us to envision, not just avoiding bad and mediocre conduct, but also merely

good conduct, and orienting ourselves instead toward realizing virtuosic conduct and relational outcomes. In short, Buddhist ethics is thus not goal or destination-oriented. It is an open-ended and improvisational art.

Buddhist ethics can, with some justification, be seen as involving elements of the virtue-based, duties-based (Deontological), and consequence-based (Utilitarian) approaches to ethics that have become dominant in Western philosophy, as well as those central to care-based (Feminist). Yet, Buddhist relational ontology distinctively orients evaluative attention away from independently existing ethical agents, patients, and actions, and toward relational quality. Moreover, while Buddhism's emphasis on relational virtuosity commits Buddhist ethics to being context-specific, it differs from Western situational ethics which evaluates actions in terms of their proximate or near-term outcomes. What matters ethically are not simply the immediate consequences of an action, but also the medium – and long-term relational ramifications of intentionally enacting specific sets of values, and their shaping of *volitional opportunities* as well as *experiential outcomes*.

In sum, Buddhist relational ontology radically expands the spatial and temporal scope of ethical evaluation. Rather than focusing on the character of both spatially and temporally “local” agents, actions, and patients, Buddhist ethics directs evaluative attention – at the very least – to show our value – and intention-enacting presences qualitatively affect the networks of internal relations that constitute us as persons in the planetary community with all other sentient beings and their sensed environments.

This expansion of evaluative scope arguably entails a metaethical commitment to fostering ethical diversity. Especially in addressing the challenges and opportunities associated with the emergence of intelligent technology; what we need is not a new “species” of ethics, but rather an ethical “ecology,” the vitality of which depends on ethical diversity or the degree to which ethical differences become the basis of a mutual contribution to shared, relational flourishing. Such an “ecological turn” in ethics is, I believe, crucial for resolving the ethical predicaments emerging as human values and intentions are innovatively scaled up by syntheses of human and machine/algorithmic intelligence.

Jewish Ethics

By Samuel J. Levine

A Jewish ethical approach, like virtually all areas of Jewish thought, is premised upon the foundational sources of Judaism, most prominently the text of the Torah, as interpreted and applied over the course of thousands of years and across geographical locations and societal settings around the world. The text of the Torah, although replete with legal precepts, includes substantial narratives as well, providing context and further meaning to the extensive legal material, while offering ethical lessons of its own. As such, Jewish

ethical responses to specific questions may involve a complex consideration of the interaction of various modes of legal and philosophical thought, as developed through millennia of teachings and traditions.

Ethical questions revolving around AI and machine learning may therefore be assessed in reference to a variety of approaches. For example, the status of AI may implicate fundamental principles underlying the nature and definition of human beings. Thus, a starting point for the analysis may very well prove to be the starting point of the Torah, the narrative portions of Genesis that depict the creation of humans as distinct from other living beings. Within Jewish tradition, some philosophers have suggested that this distinction is grounded in the human power of speech, reflecting humans' unique nature as a physical being also created in the image of God. At the same time, humans are thought to be distinct from other divine beings – the angels – in that humans have the power of free will, while angels are compelled to act in accordance with their roles as decreed by God. Whatever the explanation, by all accounts, Jewish ethics, and Jewish law value human life as paramount, thereby outweighing nearly all other concerns. The differing, yet complementary, views of the fundamental qualities of human nature may play a role in evaluating ethical approaches to AI.

Of course, the creation narrative is, by definition, the beginning of the analysis. With the passage of time, technological and societal changes impact law and ethics, as the unanticipated – and the once unimagined²² – emerge as a reality. Perhaps both acknowledging and anticipating inevitable advances in technology, Jewish sources have explored such issues as the legal status of newly-created beings created by humans and possessing quasi-human qualities, as well as the legal implications of both eternal life and resurrection from death. The emergence of AI may bring some of these issues, which have heretofore remained largely in the realm of theory, to the forefront of practical legal and ethical consideration, serving as yet another among countless examples in which ancient Jewish precepts and principles have been continuously examined, interpreted, and applied to new circumstances.

Indigenous Ethics: Actions as Interactions

By Joseph Len Miller and Andrea Sullivan-Clarke

The question of what an Indigenous moral theory looks like is difficult to answer. First, there is the problem of “Pan-Indianism.”²³ Thinking of “Indigenous” peoples as a homogenous group is problematic given the differences that exist between tribes. Second, historically, the philosophical thought of Indigenous people has been seriously underestimated. Most Western thinkers have assumed that Indigenous people were too primitive, or even “savage,” to engage in serious reflection regarding abstract topics/questions.²⁴ This history has lasting impacts on Indigenous philosophy. Not only do Indigenous ideas, even the most fundamental ones, have to be “justified”

according to Western standards, but these ideas have to be explained in a context other than the one in which they developed.²⁵

That being said, one important common focus of Indigenous ethics is the interconnectedness of all things (e.g., people, the land, non-human animals, past- and future generations, etc.).²⁶ The universe is a living entity and is understood to be in “constant transition.”²⁷ This provides Indigenous peoples a context that “operates on the principles of balance of harmony.”²⁸ People are born into community and relationships. These include nonhuman relations, such as spirits, rocks, rivers, members of non-human animal species, etc. Each entity with which we stand in relation is different, and so our actions toward our relations will also be different.²⁹ Instead of providing universal principles for guiding behavior, there are key concepts that provide a foundation and guidance for ethical decision-making. These concepts include those of harmony, reciprocity, gratitude, and humility. Understanding how these concepts relate to one another can help to better understand how to enact these concepts in different contexts. The correct way to live, and act, is then informed by what we know of our relations and our interconnectedness with these concepts.

One of the primary concepts is that of *harmony*. Harmony exists when there is a balance between exchanges and interactions with one’s surroundings.³⁰ Balance and harmony, features of the world that we are born into, offer a guide for ensuring the well-being of our relations and ourselves.³¹ Given the interdependence and relationships between all things, every interaction has an effect on the well-being of an individual and their surroundings. In other words, any action is an interaction (i.e., (inter)action). To promote balance in these interactions, an individual has to know how to *reciprocate*.³² Reciprocation can take many forms (i.e., there is not a singular “right” way to reciprocate), but it should be tailored to the being with whom one’s interacting. The aim of reciprocating is to bring balance to one’s relationships so that all beings involved can live together peacefully. In order to live together peacefully with one’s surroundings, and appropriately reciprocate, one needs to act with *love*, gratitude, and humility.

With these concepts in mind, for any particular (inter)action one should consider the following questions: What action would bring about harmony? How should I reciprocate what I have been given? Am I acting with love, gratitude, and humility? Notice, the answers to these questions are going to depend heavily on one’s surroundings and context. Answering these questions in an appropriate way requires having an intimate knowledge of one’s surroundings and relationships. For example, knowing how to bring about harmony – i.e., knowing how to reciprocate – in one’s relationship with the land requires knowing intimate details about the soil, plant life, bodies of water, weather patterns, interdependence between plants and animals in the area, etc.

Partly as a means of preserving, in the presence of settler colonialism, the ways of life that are structured around these concepts, some political concepts took on a more prominent role in Indigenous ethics. This includes the

concepts of sovereignty and reclamation. Since the focus of this collection is ethics, we have prioritized foundational ethical concepts that inform ethical decision-making in Indigenous philosophy. However, given the importance and inspiring recent publicity of interest concerning Indigenous data sovereignty,³³ we would be remiss not to share some resources that highlight how these concepts (sovereignty and reclamation) are used in the *collection and use* of data pertaining to Indigenous peoples.

Kukutai and Taylor have recently edited a volume that collects papers in support of “the inherent and inalienable rights and interests of [I]ndigenous peoples relating to the collection, ownership, and application of data about their people, lifeways and territories.”³⁴ Rodriguez-Lonebear and Martinez present an argument in support of “the repositioning of authority over Indigenous data back to Indigenous peoples.”³⁵ Carroll et al. articulate, describe, and present examples of the CARE Principles for Indigenous Data Governance (Collective Benefit, Authority to Control, Responsibility, and Ethics).³⁶

In general, Indigenous peoples approach the question of how to live well with humility because they realize that we are only a small part of the universe. We are dependent upon the well-being and generosity of our relatives (i.e., all of our relations) to survive. Dysfunction, chaos, disorder, and the deterioration of the well-being of our relatives creates disharmony and serve as an indication that our actions are misguided and we must change our ways.

Notes

1. Nachshon Goltz, John Zeleznikow, and Tracey Dowdeswell, “From the Tree of Knowledge and the Golem of Prague to Kosher Autonomous Cars: The Ethics of Artificial Intelligence Through Jewish Eyes,” *Oxford Journal of Law & Religion*, 9 (2020): 134, doi: <https://doi.org/10.1093/ojlr/rwaa015>.
2. Although it is commonly thought that utilitarianism is about the “greatest good for the greatest number of people,” Bentham and all the major Utilitarians after him have been very explicit in saying that Utilitarians must take into account the pleasures and pains of animals, and indeed of all sentient beings.
3. Fyodor Dostoevsky, *The Brothers Karamzov*, trans. Constance Garnett (Mineola, NY: Dover Publications, 2005).
4. Anderson Chuck, Wiktor Adamowicz, Philip Jacobs, Arto Ohinmaa, Bruce Dick, and Saifudin Rashid. “The Willingness to Pay for Reducing Pain and Pain-Related Disability,” *Value in Health: The Journal of the International Society for Pharmacoeconomics and Outcomes Research*, 12, no. 4 (June 2009): 498–506, doi: <https://doi.org/10.1111/j.1524-4733.2008.00457.x>.
5. Daniel Read, and George Loewenstein, “Enduring Pain for Money: Decisions Based on the Perception and Memory of Pain,” *Journal of Behavioral Decision Making*, 12, no. 1 (1999): 1–17, doi: [https://doi.org/10.1002/\(SICI\)1099-0771\(199903\)12:1<1::AID-BDM310>3.0.CO;2-V](https://doi.org/10.1002/(SICI)1099-0771(199903)12:1<1::AID-BDM310>3.0.CO;2-V).
6. Revealed preference, or sometimes called inferred preference, is not the same as true preference. Revealed preferences are therefore still proxies for what preference Utilitarians want to measure, albeit much better proxies for preference Utilitarians than they are for classical Utilitarians.

7. See e.g.: China National Health Development Research Center, Report, June 28, 2015, accessed July 2, 2022, http://ncncd.chinacdc.cn/xzzq_1/202010/W020201010848412702955.pdf; and Institute for Clinical & Economic Review, "Cost-Effectiveness, the QALY, and the evLYG," *Institute for Clinical & Economic Review* (Website), 2022, accessed July 2, 2022, <https://icer.org/our-approach/methods-process/cost-effectiveness-the-qaly-and-the-evlyg/>.
8. Vincent François, Raphael Fonteneau, and Damien Ernst, "How to Discount Deep Reinforcement Learning: Towards New Dynamic Strategies," arXiv, December 7, 2015, doi: <https://arxiv.org/abs/1512.02011>.
9. Amos Tversky, and Daniel Kahneman, "Advances in Prospect Theory: Cumulative Representation of Uncertainty," *Journal of Risk and Uncertainty*, 5, no. 4 (1 October 1992): 297–323, doi: <https://doi.org/10.1007/BF00122574>.
10. Nicholas Barberis, "The Psychology of Tail Events: Progress and Challenges," *American Economic Review*, 103, no. 3 (May 2013): 611–16, doi: <https://doi.org/10.1257/aer.103.3.611>.
11. Daniel Kahneman, and Amos Tversky, eds. *Choices, Values and Frames* (Cambridge: Cambridge University Press, 2000).
12. W. H. Desvousges, F. Reed Johnson, R. W. Dunford, Kevin J. Boyle, Sara P. Hudson, and K. Nicole Wilson, "Measuring Nonuse Damages Using Contingent Valuation: An Experimental Evaluation of Accuracy, 2nd ed.," *RTI International*, September 28, 2010, doi: <https://doi.org/10.3768/rtipress.2009.bk.0001.1009>.
13. Jonathan Z. Berman, Alixandra Barasch, Emma E. Levine, and Deborah A. Small, "Impediments to Effective Altruism: The Role of Subjective Preferences in Charitable Giving," *Psychological Science* 29, no. 5 (1 May 2018): 834–44, doi: <https://doi.org/10.1177/0956797617747648>.
14. Giving What We Can, "Maximise Your Charitable Input," *Giving What We Can* (Website), n.d., accessed July 2, 2022, <https://www.givingwhatwecan.org/>; 80,000 Hours, "You Have 80,000 Hours in Your Career," *80,000 Hours* (Website), 2021, accessed July 2, 2022, <https://80000hours.org/>; The Life You Can Save, "Not All Charities Are Equal: Some Charities Can be Hundreds or Even Thousands of Times More Impactful Than Others," *The Life You Can Save* (Website), 2022, accessed July 2, 2022, <https://www.thelifeyoucansave.org/>; Animal Charity Evaluators, "Learn How You Can Be a More Effective Advocate," Animal Charity Evaluators (Website), 2022, accessed July 2, 2022, <https://animalcharityevaluators.org/>.
15. See Rosalind Hursthouse, *On Virtue Ethics* (Oxford: Oxford University Press, 1999), p. 28.
16. A challenge raised by Bernard Williams, in his "Replies," in *World, Mind and Ethics: Essays on the Ethical Philosophy of Bernard Williams*, ed. J. Altham and R. Harrison, 185–224 (Cambridge: Cambridge University Press, 1995).
17. For a more extensive argument for this point, see: Shannon Vallor, *Technology and the Virtues: A Philosophical Guide to a Future Worth Wanting* (Oxford: Oxford University Press, 2016), Chapter 1.
18. Aristotle, *Nicomachean Ethics*, trans. C.D.C. Reeve (Indianapolis: Hackett Publishing, 2014), Book II, Chapters 6–9.
19. See: Philippa Foot, *Natural Goodness* (Oxford: Oxford University Press, 2001).
20. Gili Kliger, "Humanism and the Ends of Empire, 1945–1960," *Modern Intellectual History*, 15, no. 3 (2018): 773–800, doi: <https://doi.org/10.1017/S1479244317000282>.
21. Aristotle, *Politics*, trans. C.D.C. Reeve (Indianapolis: Hackett Publishing, 2017).
22. It is important to note that according to Jewish belief, 'There is no new under the sun' (Ecclesiastes 1:9) as stated by King Solomon; thus, whatever new technology is emerging, its equivalent, in one form or another, was already in existence in the past. An example for such an analogy is the 'Golem,' created by Jewish Kabbalists (the

- sage of the Jewish mysticism) across the generations and originating in the first man, and the Talmud. For a thorough discussion of this example, see Goltz, Nachshon, John Zeleznikow, and Tracey Dowdeswell. "From the Tree of Knowledge and the Golem of Prague to Kosher Autonomous Cars: The Ethics of Artificial Intelligence Through Jewish Eyes." *Oxford Journal of Law & Religion*, 9 (2020): 132–156. doi: <https://doi.org/10.1093/ojlr/rwaa015>.
23. Pan-Indianism is the assumption that all Indigenous thought and culture are similar enough to attribute similar concepts and beliefs to all Indigenous populations. Differences exist, not just between tribes and how they relate to different federal governments (e.g., how Indigenous peoples in United States, Canada, Australia, etc. relate to their respective federal governments), but between Indigenous peoples colonized by the same federal governments (e.g., differences exist between the Mvskoke, Diné, Luiseño, Lenape, etc. in the United States). In North America alone there are more than 1,200 First Nation and Native American communities. The worldviews of these peoples may share some similar themes, but in no way should it be assumed that an individual can speak for Indigenous peoples generally. See: Åke Hultkrantz, *Native Religions of North America* (New York: Harper & Row, 1987), 11, 27; and D. McPherson, and J. D. Rabb, *Indian from the Inside: Native American Philosophy and Cultural Renewal* (Jefferson NC: McFarland and Company, 2011), 12.
 24. For example, consider that Thomas Hobbes and John Locke both considered Indigenous nations as real-world examples of the state of nature. See: Thomas Hobbes, *Leviathan* (London, J. M. Dent & Sons, 1931); John Locke, *Two Treatises of Government* (London, 1713).
 25. Indigenous concepts and ideas have to be explained in English and explained in reference to western ideas that are taken to be foundational, agreed upon, or otherwise unquestionable. Doing this not only dilutes or changes the ideas, but the expectation that western ideas take priority, or that ideas have to be put in western terms, puts a greater intellectual burden on Indigenous peoples.
 26. Since we are both enrolled members of Muscogee Nation, our explanation and understanding of Indigenous ethics will likely be rooted in Mvskoke concepts and Mvskoke ways of life. Although we draw from other sources, these concepts and explanations are not intended to be generalized to all other Indigenous groups. Though similarities may exist, this is just an example of what an account of Indigenous ethics may look like.
 27. Viola F Cordova. *How It Is: The Native American Philosophy of V.F. Cordova*, ed. Kathleen Dean Moore, Kurt Peters, Ted Jojola, Amber Lacy, with Foreword by Linda Hogan (Tucson: University of Arizona Press, 2007), 253–4.
 28. Cordova, *How It Is*, 253–4.
 29. Brian Burkhardt, *Indigenizing Philosophy through the Land: A Trickster Methodology for Decolonizing Environmental Ethics and Indigenous Futures* (East Lansing: Michigan State University Press, 2019), 201.
 30. Jean Chaudhuri, and Joyotpaul Chaudhuri, *A Sacred Path: The Way of the Muscogee Creeks* (Los Angeles: UCLA American Indian Studies Center, 2001).
 31. Joseph Len Miller, "What Do We Need to Know to Live in Harmony with Our Surroundings?" in *Ways of Being in the World: An Introduction to Indigenous Philosophy*, ed. A. Sullivan-Clarke (Peterborough: Broadview Press, forthcoming).
 32. Chaudhuri, *A Sacred Path*.
 33. See e.g.: Mayi Kuwayu, "Indigenous Data Sovereignty Principles," *Mayi Kuwayu: The National Study of Aboriginal & Torres Strait Islander Wellbeing*, 2022, accessed July 1, 2022, <https://mkstudy.com.au/indigenousdatasovereigntyprinciples/>; United States Indigenous Data Sovereignty Network (USIDSN), "About the US Indigenous Data Sovereignty Network," *USIDSN*, 2022, accessed July 1, 2022, <https://usindigenousdata.org/about-us>, which states that the USIDSN "helps ensure that data for and about

Indigenous nations and peoples in the US... are utilized to advance Indigenous aspirations for collective and individual wellbeing”; IndigiData, “Empowering Indigenous Data Leaders,” *IndigiData: Indigenous Data Science Education*, 2022, accessed July 1, 2022, <https://indigidata.nativebio.org/>, which hold workshops for Indigenous data scientists; and S. Imbler, “Training the Next Generation of Indigenous Data Scientists,” *New York Times*, June 29, 2021, accessed July 1, 2022, <https://www.nytimes.com/2021/06/29/science/indigenous-data-microbiome-science.html>.

34. Tahu Kukutai and John Taylor, *Indigenous Data Sovereignty: Toward an Agenda* (Canberra: Australian National University Press, 2016), 2.
35. Stephanie Russo Carroll, Desi Rodriguez-Lonebear, and Andrew Martinez, “Indigenous Data Governance: Strategies from United States Native Nations,” *Data Science Journal*, 18, no. 1 (2019): 31, doi: <http://doi.org/10.5334/dsj-2019-031>.
36. Stephanie Russo Carroll, Ibrahim Garba, Oscar L. Figueroa-Rodriguez, Jarita Holbrook, Raymond Lovett, Simeon Materechera, Mark Parsons, Kay Raseroka, Desi Rodriguez-Lonebear, Robyn Rowe, *et al.*, “The CARE Principles for Indigenous Data Governance,” *Data Science Journal*, 19, no. 1 (2020): 43, doi: <http://doi.org/10.5334/dsj-2020-043>, 43; See also University of Toronto Libraries, “Research Guides: Indigenous Studies: Indigenous Data Sovereignty,” *University of Toronto Libraries*, 2022, accessed July 1, 2022, <https://guides.library.utoronto.ca/indigenoustudies/datasovereignty>; Native Nations Institute, “Indigenous Data Sovereignty and Governance,” *Native Nations Institute, University of Arizona*, 2022, accessed July 1, 2022, <https://nni.arizona.edu/programs-projects/policy-analysis-research/indigenous-data-sovereignty-and-governance>.